Source-Domain Retention: How Much Do Adapted Checkpoints Forget?

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1 Motivation and setup

Every comparison elsewhere in this study scores a checkpoint on its *target* domain – the domain the MMD term (or plain fine-tune) was meant to adapt *to*. This document asks the complementary question: after that adaptation/fine-tune, how much of the *source* domain does the checkpoint still recognize?

Every checkpoint below that reached a paper table via regime no_mmd, naive_mmd, reg_mmd, or reg_mmd_smaller_weight (cross-referenced against `experiment_summary_v6.tex`, `experiment_summary_v7.tex`, and `freeze_epoch_ablation.tex`) was re-evaluated with a plain `model.val()` call on its own source domain's held-out val split – the same split used to select `best.pt` during that domain's original pretraining run. Baseline/Solo checkpoints are excluded throughout: they train on the target domain from a COCO-direct start and never see a source domain, so retention does not apply to them. Imgsz 1920 throughout, matching training resolution.

A caveat. These tables report only the *post-adaptation* source-domain score. The natural reference point – how well the original pretrained-on-source checkpoint scored on this same val split, before it was ever repurposed for adaptation – is not yet included here; see Section 8.

2 Real→Syn direction: source = real_large

Source = real_large, target = syn_small. `reg_mmd` and `reg_mmd_smaller_weight` are the `-4` reruns (matching `experiment_summary_v7.tex`'s real→syn table, which superseded the earlier unsuffixed runs used in `experiment_summary_v6.tex`).

Model	person mAP50	person mAP50-95	vehicle mAP50	vehicle mAP50-95
no_mmd	0.020	0.007	0.350	0.213
naive_mmd	0.025	0.006	0.307	0.191
reg_mmd	0.040	0.010	0.333	0.210
reg_mmd_smaller_weight	0.030	0.007	0.319	0.199

Table 1: Evaluated on real_large val (548 images) – i.e. how well each syn_small-adapted checkpoint still detects the real domain it started from.

All four variants collapse far below the ∼0.67–0.70 person / ∼0.69–0.72 vehicle mAP50 this same source domain reaches when evaluated on itself right after pretraining (Table 1 of `experiment_summary_v6.tex`, target-domain column). Both MMD variants retain slightly more than no_mmd on every column, and reg_mmd is the best of the three MMD variants here – a small but consistent anchoring effect from the source-domain batches seen during adaptation.

3 ClearNoon source: syn_large_clearnoon

Source = syn_large_clearnoon, target = real_small. Bandwidth-freeze-epoch = 1 for the regularized variants (matching the ClearNoon side of `experiment_summary_v6.tex` Section 2.1).

Model	person mAP50	person mAP50-95	vehicle mAP50	vehicle mAP50-95
no_mmd	0.085	0.020	0.273	0.173
naive_mmd	0.108	0.026	0.287	0.177
reg_mmd	0.084	0.019	0.283	0.176
reg_mmd_smaller_weight	0.091	0.020	0.286	0.180

Table 2: Evaluated on syn_large_clearnoon val.

Here naive_mmd – the unregularized variant – retains the most source signal on both classes, ahead of both regularized variants and no_mmd. This is the one group in this study where regularizing the MMD term does not help retention.

4 All-weather synthetic source: syn_large

Source = syn_large (6400 images, all four weather conditions), target = real_small. Both freeze-epoch settings are included since both appear in `freeze_epoch_ablation.tex`.

Model	person mAP50	person mAP50-95	vehicle mAP50	vehicle mAP50-95
no_mmd	0.100	0.025	0.201	0.129
naive_mmd	0.101	0.025	0.200	0.129
reg_mmd, freeze=1	0.112	0.026	0.230	0.148
reg_mmd, freeze=5	0.125	0.032	0.219	0.142
reg_mmd_smaller_weight, freeze=1	0.120	0.030	0.227	0.146
reg_mmd_smaller_weight, freeze=5	0.109	0.028	0.222	0.143

Table 3: Evaluated on syn_large val (640 images).

Both regularized variants clearly beat no_mmd/naive_mmd on every column here – unlike the ClearNoon group above, regularization helps retention when the source domain is the full four-weather set. reg_mmd at freeze=5 gives the best person retention of the whole study (0.125/0.032); reg_mmd at freeze=1 gives the best vehicle retention (0.230/0.148). There is no single freeze-epoch setting that wins on both classes at once, echoing the person/vehicle trade-off already documented on the target-domain side in `experiment_summary_v6.tex` Section 2.2.

5 Vehicle-only source: syn_large_vehicle_only

Source = syn_large_vehicle_only, target = real_small_vehicle_only (person removed from training entirely). Bandwidth-freeze-epoch = 5.
Both regularized variants beat no_mmd and naive_mmd here, with reg_mmd slightly ahead of reg_mmd_smaller_weight – the same direction as the all-weather syn_large group above, on the one class both groups share.

6 syn_xlarge as source

Source = syn_xlarge (10240 images, all weathers), across four different targets/ scales: real_small (YOLOv10n and YOLOv10m) and real_large at two bandwidth-freeze settings.

Model	vehicle mAP50	vehicle mAP50-95
no_mmd	0.211	0.137
naive_mmd	0.216	0.140
reg_mmd	0.225	0.147
reg_mmd_smaller_weight	0.223	0.147

Table 4: Evaluated on syn_large_vehicle_only val. No person column: person was never part of this source or target dataset.

Model	person mAP50	person mAP50-95	vehicle mAP50	vehicle mAP50-95
→real_small, reg_mmd (freeze=5)	0.139	0.035	0.236	0.155
→real_small, reg_mmd, YOLOv10m (freeze=5)	0.133	0.035	0.222	0.140
→real_large, reg_mmd (freeze=6)	0.104	0.027	0.219	0.142
→real_large, reg_mmd, stronger weight (freeze=25)	0.138	0.036	0.238	0.157

Table 5: Evaluated on syn_xlarge val. No no_mmd/naive_mmd control exists yet at this source scale (see Open items).

Within the real_large-target pair, the longer, later-freezing stronger-weight variant (freeze=25, 120 epochs) retains source performance noticeably better than the freeze=6 variant on every column – the opposite of what its already-narrower target-domain gap (`experiment_summary_v7.tex` Section 3) might suggest: this variant is not just closer to solo on the target side, it also forgets its source less. The YOLOv10n vs. YOLOv10m comparison (both real_small-target, freeze=5) shows the smaller backbone retaining slightly more of the source domain on every column, despite YOLOv10m winning on the target-domain side (`experiment_summary_v7.tex` Section 3.1) – larger capacity here appears to specialize toward the target at a small additional cost to the source.

7 Overall pattern

Across all five source domains, one pattern holds without exception: every adapted or no-MMD checkpoint's source-domain score is a small fraction of what that same domain reaches when evaluated on itself immediately after pretraining – adaptation training, MMD-regularized or not, does not preserve source-domain competence. Within that collapse, MMD generally retains more than plain no-MMD fine-tuning (4 of 5 groups), and regularizing the MMD term generally retains more than the unregularized naive variant (3 of 4 groups where both exist) – but the ClearNoon source group inverts both of these directions, so neither is a universal rule.

8 Open items

- No pre-adaptation reference point: each source domain's own pretrain checkpoint, evaluated on this same val split immediately after pretraining, would turn these absolute numbers into a proper retention *delta*.
- No no_mmd/naive_mmd control at the syn_xlarge source scale, for either the real_small or real_large target – only regularized variants have been run so far at this source scale.
- person mAP50-95 in particular is often an order of magnitude below the vehicle column across every group; whether this reflects genuine source-domain person forgetting or a checkpoint-selection artifact (person's volatility already flagged in `experiment_summary_v6.tex` Section 2.2) has not been separately investigated here.